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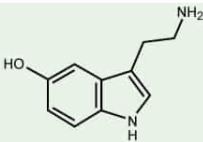
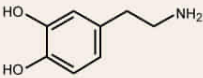
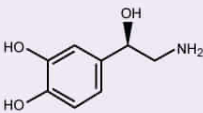
According to the monoamine hypothesis:

- ☐ A. abnormally high serum cortisol levels underlie most depressive symptoms. [4%]
- ☐ B. abnormalities in a single neurotransmitter cause most depressive symptoms. [53%]
- ☒ C. abnormalities in various neurotransmitters cause depressive symptoms. [40%]
- ☐ D. DBS would be an ineffective treatment for depressive symptoms. [2%]

Correct

40% answered correctly

Explanation:

Monoamine	Structure	Associated functions
Serotonin		Positive mood Satisfaction Social dominance
Dopamine		Pleasure Reward
Norepinephrine		Alertness Attention Memory formation

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The **monoamine hypothesis** is a biomedical model of depression. Central to this hypothesis is the idea that **depletion of monoamine neurotransmitters** in the central nervous system directly **causes depression** symptoms. The monoamines (serotonin, norepinephrine, and dopamine) are neurotransmitters that have a single amine group in their molecular structure and are associated with emotion regulation.

Serotonin is associated with a **positive mood**, feelings of satisfaction, and social dominance. In the central nervous system, **norepinephrine** enhances **alertness**, attention, and memory formation. **Dopamine** is associated with feelings of **pleasure** and **reward**. Support for the monoamine hypothesis comes from evidence that taking antidepressant medications that increase the levels of some or all of these neurotransmitters alleviates depression symptoms for many individuals.

(Choice A) Although a chronically high level of the stress hormone cortisol is associated with depression symptoms, cortisol, a glucocorticoid, is not a part of the monoamine hypothesis.

(Choice B) The monoamine hypothesis attributes depression symptoms to several monoamine neurotransmitters. It does not suggest that depression is caused by a *single* monoamine.

(Choice D) The monoamine hypothesis is supported by evidence that increasing monoamine levels in the brain alleviates depression symptoms. DBS is thought to disrupt dysfunctional patterns of electrical activity in the brain, so it is unclear how this alters monoamine levels.

Educational objective:

The monoamine hypothesis states that a depletion of monoamine neurotransmitters (serotonin, norepinephrine, and dopamine) in the central nervous system causes depression symptoms.

Time spent: 0 sec
Subject:
Behavioral Sciences

QID: 400251
System:
Sensation, Perception, and Consciousness

Major depression is a mood disorder that affects nearly 7% of the American population and is the most common cause of disability worldwide. Standard treatment typically involves antidepressant medication alone, although a combination of antidepressant medication and psychological treatment, such as psychotherapy or cognitive-behavioral therapy, is often more effective. However, because depression has genetic, environmental, and psychological contributing factors, treatment can be challenging. Approximately 10% of those suffering from depression have treatment-resistant depression (TRD) that does not respond to standard treatment.

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Which of the following brain regions is likely the most effective DBS target for a Parkinson disease patient?

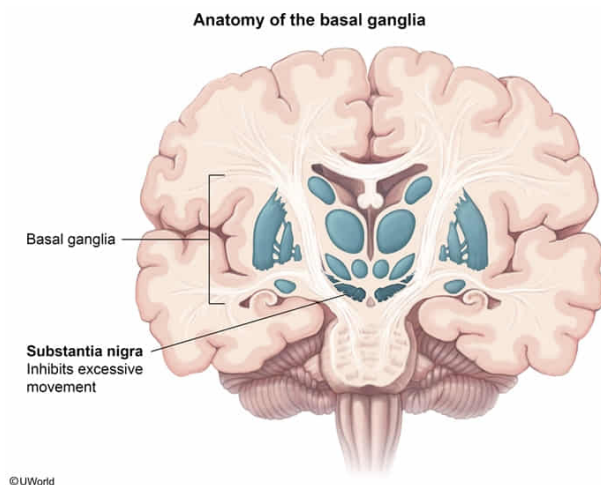
- ☐ A. Amygdala [4%]
- ☒ B. Basal ganglia [55%]
- ☐ C. Hippocampus [6%]

☐ D. Motor cortex [32%]

Correct

55% answered correctly

Explanation:



Parkinson disease is a progressive neurodegenerative disease caused by **death of dopaminergic neurons** in the **substantia nigra**, located in the midbrain (mesencephalon). The substantia nigra is part of a group of neural structures collectively known as the **basal ganglia**, which are involved in the selection and execution of voluntary motor programs.

Dopamine has several functions in the central nervous system, including **mediating the reward pathway** and **motor control**. The basal ganglia are composed of three dopaminergic pathways: the direct pathway, the indirect pathway, and the nigrostriatal pathway. The direct and indirect pathways have excitatory and inhibitory effects on motor function, respectively, and the nigrostriatal pathway helps maintain the balance between these two pathways. The nigrostriatal pathway is damaged when dopaminergic neurons in the substantia nigra are destroyed. Therefore, the balance between excitatory and inhibitory motor commands deteriorates.

Resulting Parkinson symptoms include various motor abnormalities: resting tremor (usually in the upper extremities), muscle rigidity, slowed movement (bradykinesia), shuffling gait, and lack of coordination. DBS in portions of the basal ganglia has been shown to relieve some of these motor symptoms.

(Choice A) The amygdala is a neural structure that is part of the limbic system and is involved in emotion regulation and learning (especially fear-based conditioning).

(Choice C) The hippocampus is a neural structure that is part of the limbic system and is involved in memory formation, learning, and recall.

(Choice D) The motor cortex sends motor commands from the brain to the muscles of the body. DBS of the motor cortex would likely cause disruptions in movement.

Educational objective:

Parkinson disease is a progressive neurodegenerative disease that involves the loss of dopaminergic neurons in the substantia nigra of the basal ganglia, resulting in motor abnormalities (resting tremor, muscle rigidity, bradykinesia, shuffling gait, lack of coordination) and symptoms of depression.

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Behavioral Sciences

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Which of the following conclusions is best supported by Table 1?

- ✔ ☒ A. No comparisons can be made between targets that used different depression measures. [34%]
- ☐ B. Stimulation of the medial forebrain bundle was the most effective treatment overall. [31%]
- ☐ C. Stimulation of the nucleus accumbens and the subcallosal cingulate gyrus was equally effective. [28%]
- ☐ D. Stimulation of the ventral capsule/ventral striatum was the least effective long-term treatment. [5%]

Explanation:

Target	Studies included	Differing depression scales and remission criteria
Nucleus accumbens	3	28-item HDRS; remission = HDRS-28 score <10
Subcallosal cingulate gyrus	6	17-item HDRS; remission = HDRS-17 score <8
Ventral capsule/ventral striatum	2	24-item HDRS; remission = HDRS-24 score <10
Medial forebrain bundle	2	MADRS; remission = MADRS score <10

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Table 1 presents data from a **meta-analysis**, a systematic, quantitative review of **multiple studies** that have attempted to answer a similar question using **comparable methodology** and **outcome measures**. By statistically **analyzing** the aggregate **data**, a meta-analysis provides a broader perspective than any single experiment.

Table 1 shows outcomes of DBS for specific brain regions. For each brain region, only studies using the same outcome criteria were pooled in the analysis. Superscripts indicate the depression scales and remission criteria used for each brain target. For example, this meta-analysis included three studies using the nucleus accumbens as a DBS target, all of which measured depression symptoms with a 28-item HDRS.

Different measures (eg, different HDRS and MADRS scales) and *different remission criteria* (eg, scores below 8 or 10) were used for different brain targets. Therefore, it is not possible to make comparisons *between* brain targets in terms of statistical significance or efficacy.

(Choice B) The two medial forebrain bundle studies used the MADRS to measure symptoms. The rest of the studies used different HDRS scales, which cannot be compared to the MADRS, so it is not possible to conclude that this target was the most effective.

(Choice C) The three nucleus accumbens studies and six subcallosal cingulate gyrus studies used different HDRS scales (28-item vs. 17-item scale) and qualifications for remission scores (<10 vs. <8). Therefore, it is not possible to compare the outcomes of these studies or conclude that DBS of these targets was equally effective.

(Choice D) The two ventral capsule/ventral striatum studies used the 24-item HDRS scale, which differs from the scales used for the other brain targets, so it is not possible to conclude that this target was the least effective.

Educational objective:

A meta-analysis statistically analyzes data combined from multiple studies with a common experimental goal. To combine study results for a meta-analysis, studies must use parallel methodology and outcome measures. Averages between groups cannot be compared if they are not based on the same outcome measures.

Time spent: 0 sec**Subject:**
Behavioral Sciences**QID:** 400253**System:**
Sensation, Perception, and Consciousness

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Stimulation of the nucleus accumbens would be expected to alleviate which of the following symptoms of depression?

- ✔ ☒ A. Anhedonia [49%]
☐ B. Difficulty concentrating [23%]
☐ C. Fatigue [17%]

☐ D. Insomnia [9%]

Correct

50% answered correctly

Explanation:

Symptoms of depression

- Feelings of sadness, hopelessness, or emptiness
- Lack of interest or pleasure
- Changes in appetite (eating more or less)
- Changes in sleep (sleeping more or less)
- Fatigue
- Low self-worth or guilt
- Cognitive difficulties
- Thoughts about death or suicidal ideation

Depression is a **mood disorder** that is characterized by feelings of sadness, hopelessness, helplessness, low self-worth and self-esteem, fatigue, lack of interest in previously enjoyed activities, loss of motivation, changes in sleep patterns (sleeping more or sleeping less), changes in appetite (eating more or eating less), cognitive difficulties (concentration, decision making, memory), body aches, and lack of pleasure (anhedonia). Those suffering from major depression may also have suicidal ideation or commit suicide.

The **nucleus accumbens** is a structure that is part of the **reward pathway** in the brain. The **reward pathway** also includes the ventral tegmental area (which produces dopamine) and portions of the prefrontal cortex. DBS of these structures could help alleviate symptoms of depression, as activation of the reward pathway is associated with feelings of pleasure and the reinforcement of rewarding behaviors.

(Choices B, C, and D) Because the nucleus accumbens is part of the reward pathway in the brain, DBS of this area is most likely to impact anhedonia and less likely to impact the cognitive (difficulty concentrating), alertness (fatigue), and sleep (insomnia) symptoms often associated with depression.

Educational objective:

Depression is a mood disorder characterized by emotional and physical symptoms, such as sadness, hopelessness, anhedonia, difficulty concentrating, fatigue, body aches, and changes in sleep and appetite. Stimulation of the nucleus accumbens in the brain's reward pathway may help alleviate anhedonia, allowing depressed individuals to feel pleasure again.

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Behavioral Sciences

QID:400254
System:
Sensation, Perception, and Consciousness

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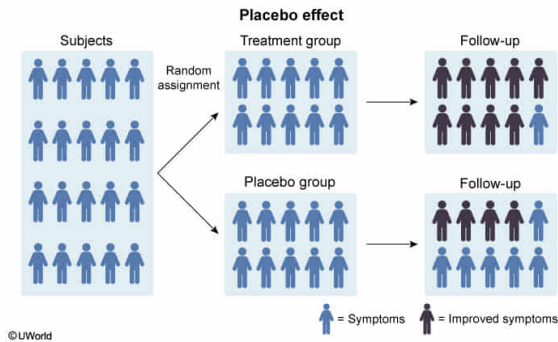
Which of the following is an inherent limitation in assessing the efficacy of DBS for controlling depression symptoms?

- ✔ ☒ A. A placebo control group would be unethical. [41%]
- ☐ B. Depressed patients are a vulnerable population. [10%]
- ☐ C. The benefits do not outweigh the risks of the procedure. [6%]
- ☐ D. The mechanism of action of DBS is unknown. [41%]

Correct

41% answered correctly

Explanation:



The **placebo effect** occurs when the administration of an **inactive substance** or **sham procedure** (the **placebo**) corresponds to the improvement of symptoms, often related to a person's belief that a treatment will work. In a placebo-controlled randomized trial, participants are randomized into either an active treatment group or a group that receives a placebo. Comparison of treatment and placebo groups helps to determine the **true benefit** of a given treatment when patients are unaware of whether they have received the real treatment or the placebo.

A placebo control group for this experiment would receive a sham surgery, with all of the same procedures as the treatment groups (including the insertion of electrodes into the brain), but without the electrical impulse. Exposing patients to the risks associated with a surgical procedure (eg, infection, damage to brain tissue) for an inactive sham procedure would be unethical.

In such cases, an active control group (in which the patient receives a standard approved treatment) is recommended for comparison.

(Choice B) Depressed patients could be a vulnerable population, depending on the severity of their disorder. However, this is not an inherent limitation of the study because depressed patients are required for researchers to determine if the treatment is effective for depression.

(Choice C) Whether the benefits would outweigh the risks of the procedure depends on the severity of the patient's symptoms and the consequences of not receiving adequate treatment. The benefits and risks of DBS would be assessed in controlled studies.

(Choice D) Although the mechanism of action of DBS is not fully understood, this is not an inherent limitation that would prevent an assessment of its efficacy.

Educational objective:

A placebo is an inactive substance or sham procedure used in an experiment as a comparison to the real treatment. Although placebo groups are often helpful in assessing the true benefit of a treatment, a placebo group may be unethical if a sham procedure could result in harm.

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Behavioral Sciences

QID: 400255

System:

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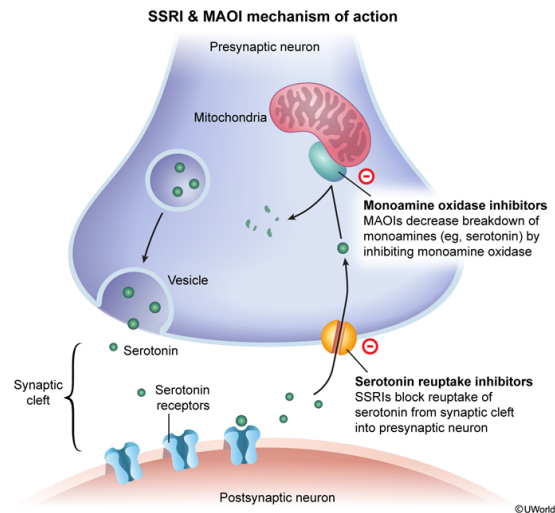
Which of the following is true regarding two classes of drugs commonly used to treat depression, selective serotonin reuptake inhibitors (SSRIs) and monoamine oxidase inhibitors (MAOIs)?

- ☐ A. Both classes of drugs stimulate the release of serotonin into the synaptic cleft. [14%]
- ☐ B. SSRIs block uptake of serotonin into the post-synaptic neuron. [25%]
- ☒ C. MAOIs decrease the breakdown of serotonin within the pre-synaptic neuron. [54%]
- ☐ D. MAOIs promote uptake of serotonin into the pre-synaptic neuron. [5%]

Correct

53% answered correctly

Explanation:



Depression symptoms appear to be related to decreased levels of the monoamine neurotransmitters serotonin, dopamine, and norepinephrine in the central nervous system (monoamine hypothesis). Many **antidepressant medications** increase the concentration or prolong the function of one or more of these neurotransmitters at the synaptic cleft. These drugs usually function via one of four general mechanisms:

1. **Increasing production** of neurotransmitters within the presynaptic neuron
2. **Promoting release** of neurotransmitters from the axon terminal into the synaptic cleft
3. **Blocking reabsorption (reuptake)** of neurotransmitters into the presynaptic neuron
4. **Decreasing breakdown** of neurotransmitters within the presynaptic neuron

For instance, MAOIs inhibit **monoamine oxidase**, an enzyme attached to the mitochondria in axon terminals that **catalyzes** the **oxidation** (breakdown) of **monoamines** such as serotonin, dopamine, and norepinephrine. This enzyme functions in the presynaptic neuron to recycle monoamines that have been removed from the synaptic cleft. Inhibition of monoamine oxidase therefore increases the concentrations of monoamines in the axon terminal and eventually within the synaptic cleft.

SSRIs belong to another class of antidepressants that selectively block the reabsorption (reuptake) of serotonin into the presynaptic neuron, thereby prolonging the presence of serotonin in the synaptic cleft.

(Choice A) Neither SSRIs nor MAOIs directly stimulate the release of serotonin or other monoamines. However, some drugs, such as serotonin-releasing agents and amphetamines, promote the release of neurotransmitters to increase the activity of these in the synaptic cleft.

(Choice B) Reuptake inhibitors such as SSRIs block the uptake of neurotransmitters into the *pre-synaptic* cleft, where they were originally synthesized and released.

(Choice D) Promoting uptake of serotonin would decrease the concentration of serotonin in the synaptic cleft, likely exacerbating the symptoms of depression.

Educational objective:

Antidepressants increase the function of one or more neurotransmitters within the synaptic cleft. This is often done by increasing the production of neurotransmitters, promoting their release, blocking their reuptake, or decreasing their breakdown within the presynaptic neuron. One class of drugs, MAOIs, inhibits the enzyme monoamine oxidase, decreasing the breakdown of serotonin, dopamine, and epinephrine.

Time spent:0 sec

Subject:

Behavioral Sciences

QID:400256

System:

Sensation, Perception, and Consciousness